

Part 1: System Design — Text-Based Chatbot for Elderly Asset Management

a. Clarifying Questions for Business Analyst

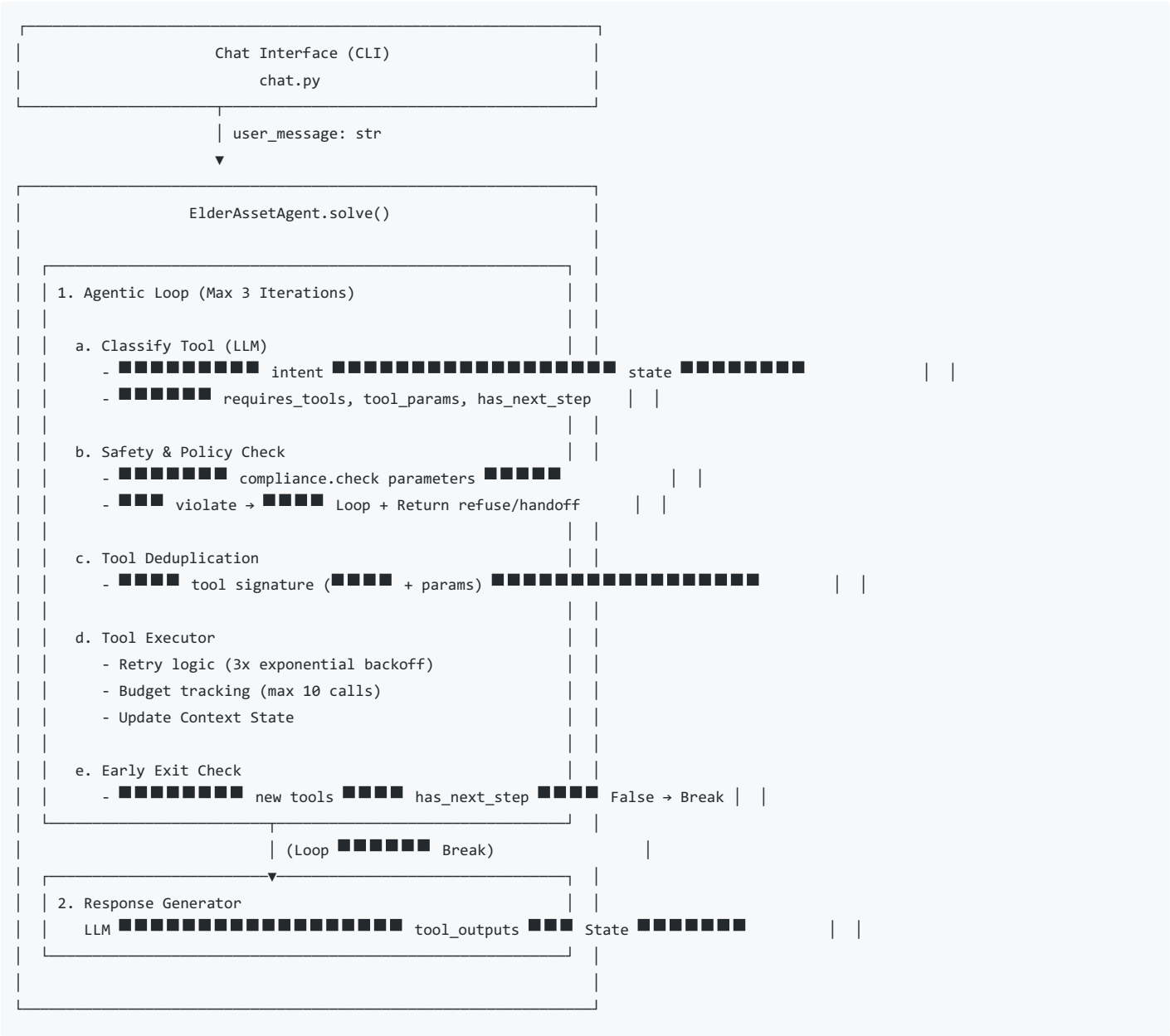
1. [Redacted] : [Redacted]
2. [Redacted] ? (Handoff): [Redacted]
3. [Redacted] : [Redacted] ([Redacted] Line FB) [Redacted]
4. [Redacted] : [Redacted] ?
- [Redacted] (Concerns & Risks)
- [Redacted] → [Redacted]
 - LLM [Redacted] → [Redacted]
tools/APIs [Redacted]
 - [Redacted] AI [Redacted] → [Redacted]

b. Design Considerations

[Redacted]	[Redacted]	[Redacted] Design
Usability ([Redacted] [Redacted] [Redacted])	[Redacted]	[Redacted] LLM [Redacted] response prompt instructions [Redacted] simplicity
Safety ([Redacted] [Redacted])	[Redacted] action [Redacted] [Redacted] / [Redacted] compliance check [Redacted]	[Redacted] Safety Engine [Redacted] hard gate [Redacted] sensitive action
Privacy ([Redacted] [Redacted] [Redacted])	[Redacted] PII [Redacted] LLM [Redacted]	Sanitize data [Redacted] LLM, [Redacted] tool [Redacted] structured data
Data Grounding ([Redacted] [Redacted])	[Redacted] tool [Redacted] LLM [Redacted]	[Redacted] response [Redacted] evidence IDs [Redacted] tool calls
Robustness ([Redacted] [Redacted])	Tools [Redacted] timeout [Redacted] return [Redacted] [Redacted] duplicates	[Redacted] retry logic, deduplication
Auditability ([Redacted] [Redacted])	[Redacted]	tool_trace [Redacted] tool call [Redacted] input/output
Multi-language ([Redacted] [Redacted])	[Redacted] input [Redacted]	LLM instruction: match user language
Scalability ([Redacted] [Redacted])	[Redacted]	Tool call budget ≤ 10 per request, efficient prompt design

c. System Architecture

Architecture Diagram



Key Components and Responsibilities

Component	Responsibility
Chat Interface (chat.py)	Input response format
Agentic Loop	ReAct (Reasoning and Acting) Tool (infinite loop)
Tool Classifier	LLM Context Tools Step (has_next_step)
Safety & Compliance Gate	parameter compliance.check LLM Policy (Fail-safe)
Tool Deduplication	Tool Signature (tool_name + json params) Token
Tool Executor	Wrapper tools 7 retry (3 exponential backoff), budget tracking (max 10 calls), trace logging

Response Component Generator	LLM tool outputs Responsibility: extract evidence IDs
LLM Client	Interface LLM provider (Gemini) – JSON parsing, Configurable temperature

Data Flow

1. User → Agent: text
2. Agent → Tool Classifier (LLM): 1 Tool
3. Agent → Safety: compliance compliance.check
4. Agent → Tools: Tool Tools
5. Agent → Loop: State Tool Classifier (LLM) 2 (has_next_step False 3)
6. Agent → Response Generator (LLM): response State
7. Agent → User: + evidence + safety info

d. Measurement & Evaluation

Metrics

Metric	
Task Completion Rate	requests status="success" ÷ total requests > 85%
Accuracy	response tool output (evidence verification) 100% (zero tolerance)
Safety Compliance	high-impact requests handoff/refuse (false negative) 100%
Handoff Rate	% requests escalate human agent 10-20% (= agent actiokn)
Response Latency	user message agent response < 10 (requests transaction)
Tool Call Efficiency	tool calls request < 5 (budget 10)
User Satisfaction	Feedback survey (1-5 rating) > 4.0

Warning Signals

Signal	
Accuracy < 100%	LLM hallucinate → evidence pipeline, validation
Handoff Rate > 30%	Agent requests → capabilities, intent classification
Safety Violations	Policy enforcement → compliance integration, test cases
Response Latency > 15	→ optimize LLM calls, cache frequent queries
	UX → response clarity

Metrics

1. Weekly Review: dashboard metrics

2. Monthly Tuning: [REDACTED] prompts, thresholds, fraud keywords [REDACTED] data [REDACTED]
3. Quarterly Evaluation: [REDACTED] scenarios [REDACTED], [REDACTED] test case

e. Safety & Action Gating

Actions [REDACTED] Gate

Action	[REDACTED]	[REDACTED]
view_balance, view_transactions, view_portfolio	Read-only	[REDACTED] ([REDACTED] compliance check)
transfer_funds	High-impact	[REDACTED] Chatbot [REDACTED] → Handoff + compliance.check()
withdraw_funds	High-impact	[REDACTED] Chatbot [REDACTED] → Handoff + compliance.check()
buy_security, sell_security	High-impact	[REDACTED] Chatbot [REDACTED] → Handoff + compliance.check()
change_beneficiary	Critical	[REDACTED] Handoff + verbal confirmation required
close_account	Critical	[REDACTED] Handoff + verbal confirmation + cooling off
update_profile	Sensitive	[REDACTED] Handoff + email + SMS confirmation
Crypto/digital asset inquiry	Fraud alert	[REDACTED] Refuse + alert fraud team

[REDACTED] Unsafe Actions [REDACTED] Architecture

1. Structural Impossibility ([REDACTED] tool [REDACTED])

- Chatbot [REDACTED] tool [REDACTED] ([REDACTED] production [REDACTED])
- Tools [REDACTED] read-only ([REDACTED]) [REDACTED] escalation ([REDACTED] support case)
- [REDACTED]: [REDACTED] LLM [REDACTED] [REDACTED] [REDACTED] tool

2. Hard-coded Compliance Gate ([REDACTED] LLM)

- Safety Engine [REDACTED] parameters [REDACTED] LLM [REDACTED] compliance.check
- [REDACTED] compliance service [REDACTED] [REDACTED] Agentic Loop [REDACTED] Tool [REDACTED]

3. Tool Call Budget

- [REDACTED] tool calls [REDACTED] 10 [REDACTED] request
- [REDACTED] infinite loops [REDACTED] excessive resource usage
- [REDACTED] budget → degrade gracefully

4. Response Schema Validation

- Output [REDACTED] agent [REDACTED] fixed schema (status, message, tool_trace, evidence, safety)
- status [REDACTED] set [REDACTED] – [REDACTED] free-form action
- [REDACTED] downstream systems ([REDACTED] audit log, monitoring) [REDACTED]

5. Fraud Detection Layer

- [REDACTED] keywords [REDACTED] fraud patterns:
 - Cryptocurrency / digital assets
 - [REDACTED]
 - [REDACTED]
 - [REDACTED]
- [REDACTED] → [REDACTED] ([REDACTED])

6. Policy Conflict Resolution

- [REDACTED] policy 2 versions (v1 [REDACTED] v2) – [REDACTED]
- [REDACTED]: v1 = [REDACTED] 60+, v2 = 65+ → [REDACTED] 60+ ([REDACTED])
- Compliance tool (compliance.check()) [REDACTED] source of truth [REDACTED] enforcement [REDACTED]